

**Course Code** : 2104CS352**Date** : 14-10-2022**Course Name** : Internet of Things**Duration** : 150 Minutes**Total Marks** : 70**Instructions:**

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

**Q.1 (A)** List the functional blocks of IoT in detail. **4**

**(B)** Define the following Terms: **3**

- 1) IoT
- 2) Devices In IoT
- 3) Data in IoT

**OR**

Recall Full Form of Terms: **3**

- 1) UART
- 2) SPI
- 3) USB

**(C)** Draw and Explain 7 layer architecture of IoT. **7**

**OR**

Explain in detail: Application of IoT in Smart cities and Smart Home.

**Q.2 (A)** Match the features of Micro-controller and Micro-processor. **4**

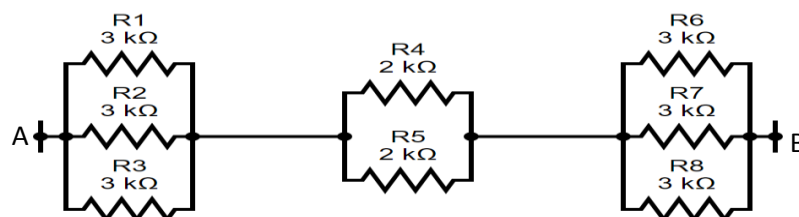
**(B)** Write down ohm's law for electrical circuit. **3**

**OR**

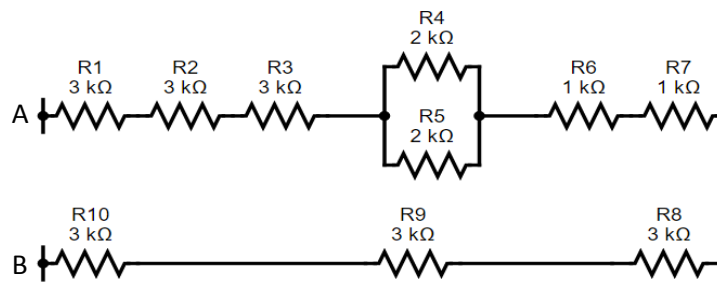
List down the SI unit of **6**

- 1) Voltage
- 2) Current
- 3) Power
- 4) Resistance
- 5) Capacitance
- 6) Inductance

**(C)** Find equivalent Resistance between node A and B. **7**

**OR**

Find equivalent Resistance between node A and B.



**Q.3 (A)** Define Servo motor with pin configuration. **4**

**(B)** Explain actuators and review functionality of Relay. **3**

**OR**

Explain sensor and review functionality of obstacle sensor.

**(C)** Write an Arduino code to measure the light intensity in the room. **7**

**OR**

Write an Arduino code for the detect obstacle. Turn On the LED when obstacle detected, else keep LED Turn OFF.

**Q.4 (A)** Demonstrate serial communication with example in Arduino IDE. **4**

**(B)** Explain an Arduino IDE function to read and write analog value to Arduino pin. **3**

**OR**

Explain the digitalWrite ( ) and digitalWrite ( ).

**(C)** Write an Arduino code to control the brightness of LED (Pin 10) as per the Input Voltage from Potentiometer (Analog Pin A5). (Use map() ) **7**

**OR**

List Analog port and digital port in Arduino UNO.

**Q.5 (A)** Explain appropriate cloud service with the help of Blynk cloud example. **4**

**(B)** Define Fog computing. **3**

**OR**

Identify the importance of Cloud computing over local computing in IoT.

**(C)** Use the Blynk library function to write the value to virtual-pin V0 and virtual pin V1. **7**

**OR**

Use the Blynk library function to read the value from virtual-pin V2 and virtual pin V3.

\*\*\*